P-S-863F
June 11, 1976
SUPERSEDING
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June 15, 1968

FEDERAL SPECIFICATION

SWEEPING COMPOUND

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers three types of sweeping compounds (see 6.1).

1.2 Classification. The sweeping compounds covered by this specification shall be of the following types as specified (see 6.2).

Type I - Sawdust-sand-mineral oil.
Type II - Sawdust-sand-water wax emulsion.
Type III - Sawdust-wax base.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-P-45 - Packaging of Bulk Quantities of Soaps, Detergents And Related Products (Chip, Powdered, Or Granular Form) For Domestic And Overseas Shipment.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables For Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American Society for Testing and Materials (ASTM) Standards:

D 93 - Flash Point by Pensky-Martens Closed Tester.
D 460 - Sampling and Chemical Analysis of Soaps and Soap Products.
E 11 - Wire-Cloth Sieves for Testing Purposes.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. The sweeping compound shall be a clean, homogeneous mixture free from lumps and caking.

3.2 Basic requirements. The sweeping compound shall conform to the requirements shown in table I when tested as specified in table II.

TABLE I. Basic requirements

Requirement	Type I		Type II		Type III	
	Min. %	Max. %	Min. %	Max. %	Min. %	Max. %
Volatile matter at 105°C., weight	--	12	--	12	--	12
Extractable oil (type I)						
(a) Applicable to product made of hard-wood sawdust	15	20	--	--	--	--
(b) Applicable to product made of soft-wood sawdust	18	26	--	--	--	--
Acid number of extracted oil (mg KOH per gm of oil)	--	17 units	--	--	--	--
Waxes and emulsifiers weight and form 1/ (types II and III)	--	--	15	20	30	45
Sand or substitute weight retained on No. 20 sieve, weight based on total sand content	35	45	40	50	--	--
Sawdust, weight retained on No. 8 sieve, weight based on total sawdust content	--	1.0	--	1.0	--	1.0
passing through No. 40 sieve, weight based on total sawdust content	22	--	18	--	43	70
Ash content weight	--	1.0	--	1.0	--	1.0
Salt, (sodium chloride) weight	--	40.0	--	40.0	--	40.0
	--	--	--	--	--	3.0
	--	0.2	--	0.2	--	0.2

1/ The chloroform extract (see 4.4.2.4) shall be in solid form at a temperature below 90° F.

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3.3 Odor. The sweeping compound shall not have a strong or rancid odor.

3.4 Sand quality. The sweeping compound shall show no visual evidence of clay pieces.

3.5 Color. The product may be the natural color or furnished with an artificial color of a light shade.

3.6 Substitution material.

3.6.1 Sand substitution. Feldspar or moulders sand may be used in lieu of the sand normally used in this product.

3.6.2 Sawdust substitution. Sisal fiber cuttings or ground excelsior may be substituted entirely or partially for the sawdust. The ground excelsior shall conform to the sieve test in table I except the weight retained on the No. 8 sieve shall be a maximum of not more than 10 percent.

3.7 Staining. The types II and III sweeping compound shall not stain or spot white bond paper when tested as specified in 4.4.2.7.

3.8 Flash point. When tested as specified in 4.4.2.8, the product's flash point shall be a minimum of 125°C.

3.9 Combustibility. When tested as specified in 4.4.2.9, the product shall not be permanently ignited and there shall be no appreciable spreading of flames on the product beyond the vicinity of the source of ignition.

3.10 Spontaneous heating. The sweeping compound shall not show evidence of spontaneous heating when tested as specified in 4.4.2.10.

3.11 Label. The container label shall include instructions for use stating the amount of the product recommended per floor area. Labels on types I and II sweeping compound are also required to state the types of floors on which the product may or may not be used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection sampling. Sampling for the inspection procedures set forth in this specification shall be performed in accordance with MIL-STD-105. The lots to be inspected shall consist of no more than one type (see 1.2) and one container size (see 5.1).

4.3 Examination.

4.3.1 Examination of preparation for delivery.

4.3.1.1 Sampling plan. The lot size for this examination shall be defined as the number of shipping containers offered for inspection at one time. The sample unit shall be defined as one such shipping container fully marked and prepared for delivery. The samples shall be selected in accordance with inspection level S-2 and the AQL shall be 2.5 percent defective.

4.3.1.2 Examination procedures. The examination shall be made to determine the compliance of the packing and marking requirements of section 5. The sample units shall be examined for the defects described below and scored accordingly.

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Examination of Preparation for Delivery	
Examine	Defects
Packing	Not as specified (see 5.1).
Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application (see 5.1).
Materials	Component missing or damaged; incomplete fill.
Workmanship	Bulging or distortion of the packing container or its component materials.

4.3.2 Examination of the filled unit container.

4.3.2.1 Sampling plan. The lot size for this examination shall be defined as the number of filled containers that contain the material manufactured within a 24 hour period in a single operation at one location using the same process and raw materials. The sample unit shall be defined as one such filled container. The sample units shall be selected in accordance with level S-2 and the AQL shall be 2.5 percent defective.

4.3.2.2 Examination procedure. The examination shall be made to determine the compliance of the marking and contents with the requirements of 3.1, 3.3, 3.4, 3.5, and 3.11. The sample units shall be examined for the defects described below and scored accordingly.

Examination of the Filled Unit Container	
Examine	Defect
Material	Not homogeneous; lumpy or caked.
Odor	Strong or rancid.
Sand quality	Evidence of clay.
Label	Not complete (see 3.11).

4.4 Testing.

4.4.1 Sampling for testing. The lot size for testing shall be defined as the number of filled containers that contain the material manufactured within a 24 hour period in a single operation at one location using the same process and raw materials. The sample unit shall be defined as one such filled container. A two pound composite shall be made by mixing equal portions of each of the sample units taken from the lot in accordance with inspection level S-1. The lot shall be unacceptable if the composite sample fails to pass any of the tests described in 4.4.2.

4.4.2 Testing procedures. The composite sample shall be tested in accordance with the instructions in table II.

TABLE II. Instructions for testing the end item

Characteristic	Spec. reference		Number of determinations per sample unit 1/	Results reported as	
	Requirement	Test Method		Pass or fail 2/	Numerically to nearest
Volatile matter at 105°C.	Table I	4.4.2.1	Average of 2		0.1 percent
Extractable oil (type I) weight	Table I	4.4.2.2	Average of 2		0.1 percent
Acid number (type I)	Table I	4.4.2.3	Average of 2		0.1

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TABLE II. Instructions for testing the end item (cont.)

Characteristic	Spec. reference Require-ment	Spec. reference Test Method	Number of determinations per sample unit 1/	Results reported as Pass or fail 2/	Results reported as Numerically to nearest
Waxes and emulsifiers:					
Weight (types II and III)	Table I	4.4.2.4	Average of 2		0.1 percent
Sand or substitute (types I and II): retained on No. 20 sieve	Table I	4.4.2.5	Average of 2		0.1 percent
Sawdust:					
retained on No. 8 sieve	Table I	4.4.2.5	Average of 2		0.1 percent
passing thru No. 40 sieve	Table I	4.4.2.5	Average of 2		0.1 percent
Ash, sawdust and sand content	Table I	4.4.2.5	Average of 2		0.1 percent
Salt (sodium chloride)	Table I	4.4.2.6	Average of 2		0.1 percent
Staining (types II and III)	3.7	4.4.2.7	1	X	
Flash Point	3.8	4.4.2.8	1		Degree
Combustibility	3.9	4.4.2.9	1	X	
Spontaneous Heating	3.10	4.4.2.10	1	X	

1/ Test reports shall include all values on which average results are based.
2/ If failure is indicated, report either description of failure or numerical point of failure as applicable.

4.4.2.1 Matter volatile at 105°C. Weigh 35 grams of the sample in a tared weighing bottle and heat with stopper removed for 2 hours in an oven at a temperature of 105° ± 2°C. Insert stopper, cool in desiccator, and weigh. Repeat to a weight constant to within 0.1 grams. Calculate loss in weight as volatile matter. Save the dried sample for the extractable oil determination in 4.4.2.2, and for the waxes and emulsifiers determination in 4.4.2.4 when applicable.

4.4.2.2 Extractable oil.

4.4.2.2.1 Apparatus required.

(1) Extraction apparatus, soxhlet, with standard tapered ground joints to fit a 250 ml. extraction flask.
(2) Fat free extraction thimbles - Whatman.
    (a) Inside diameter - 33 millimeters.
    (b) Inside length - 94 millimeters.

4.4.2.2.2 Procedure. Transfer quantitatively the dried sample from 4.4.2.1 to an extraction thimble. Pour 150 milliliters of ethyl ether into a tared 250 milliliter extraction flask. Assemble the extraction flask tube and condenser on a steam bath and extract for 10 cycles. Stop extraction process so as to retain the maximum volume of ether in the extraction flask. Connect extraction flask to a distilling condenser and distill off the ether using a steam bath. Remove the last traces of ether from the flask on the steam bath. Cool the flask in a desiccator, weigh, and calculate the percentage of oil. Save the air-dried thimble residue for the sieve analysis in 4.4.2.5. Save the oil for the acid number determination in 4.4.2.3.

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4.4.2.3 Acid number. Transfer 2 to 3 grams of the oil obtained in 4.4.2.2 to a tared Erlenmeyer flask and weigh. Add 30 milliliters of a 1:1 mixture of 95 percent ethyl alcohol and benzol which has been neutralized to phenolphthalein with alkali. Heat on the steam bath to boiling and swirl the flask. Add phenolphthalein and titrate with standardized 0.5 N KOH to a faint pink color that persists for 30 seconds. Calculate the acid number as follows:

4.4.2.3.1 Calculation.

$$\text{Acid number} = \frac{(a)\ (b)\ (56.1)}{c}$$

Where:

a = milliliters of standard KOH
b = normality of standard alkali solution
c = grams of oil

4.4.2.4 Waxes, emulsifiers, and form, types II and III. The apparatus and procedure and sample size specified in 4.4.2.2 shall be used, except that chloroform shall be the extracting solvent instead of ether. Examine the chloroform extract at a temperature of 90°F. for form. Save the thimble residue for the sieve analysis in 4.4.2.5.

4.4.2.5 Sieve test.

4.4.2.5.1 Apparatus. Tests shall be made using a single-eccentric type mechanical shaker which imparts to the sieves a rotary motion and tapping action of a uniform speed of 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. (Rotap sieve shaker is satisfactory.) Sieves shall conform to ASTM E 11.

4.4.2.5.2 Sieving procedure. Transfer the thimble residue from 4.4.2.2 or 4.4.2.4, depending on the type of product, to the coarser screen of the nest of appropriate sieves and screen in shaking apparatus. Continue screening for 5 minute intervals until the weight retained on the finer screen is constant. Weigh the material on the sieves and that which passed through the sieves. Save all the material for ash, sawdust and sand content in 4.4.2.5.3.

4.4.2.5.3 Ashing procedure. Ash the material collected on each sieve separately and that which passed through the sieves. Ash the material to constant weight, cool in a desiccator, and record the weight of the ash in each sample.

4.4.2.5.4 Calculations. The percent total sand for types I and II and the percent total ash for type III are the combined weight of the four subsamples after ashing divided by the total weight of the sample before sieving. The sawdust content of the total sample and each of the subsamples is the loss in weight during ashing. The percent total sawdust is the combined weight loss of the four subsamples divided by the sample weight before sieving. The percent sand retained on the No. 20 sieve is the weight of the subsample retained on the sieve after ashing divided by the total weight of ash. The percent sawdust retained on the No. 8 sieve and the percent sawdust passing through the No. 40 sieve is the loss in weight of the applicable subsample during ashing divided by the combined weight loss of the sieve samples during ashing.

4.4.2.6 Salt (sodium chloride). The salt content of the sweeping compound shall be determined in accordance with ASTM D 460.

4.4.2.7 Staining. Cover a sheet of white bond paper with sweeping compound to a depth of 1/2 inch. Let stand for one-half hour at a temperature of 70° to 80°F (20° to 27°C). Remove the compound and examine for spotting or staining of the paper.

4.4.2.8 Flash point. The flash point apparatus used in this test shall be as specified in ASTM D 93.

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The cup of the flash point apparatus is filled with the product to the level indicated by the annular mark. The cover is placed on the cup, and the latter set in the heating chamber. The thermometer is inserted through the cover and into the product. The test flame is lighted and adjusted so that it is the size of a bead 5/32 inch (3.97 millimeters) in diameter.

Heat is supplied to the apparatus at such a rate that the temperature read on the thermometer inserted in the product increases at 10°F ± 1°F (7 ± 1°C) per minute. Application of the test flame is made at each temperature reading which is a multiple of 5°F (3°C). The first application of the test flame is made at a temperature at least 30°F (17°C) below the actual flash point. Application of the test flame is made by operating the device controlling the shutter and the test flame burner so that the flame is lowered in 0.5 second, remains in the lowered position for one second, and is quickly raised to its high position.

The flash point is taken as the temperature read on the thermometer (inserted in the sample) at the time of the flame application that causes a distinct flash in the interior of the cup.

4.4.2.9 Combustibility. A sample of the product, previously conditioned to constant weight in air at ordinary room temperature and 40 percent relative humidity, is piled loosely in the form of a cone, with a base six inches in diameter and an altitude of two inches. A gas test flame, 1/2 inch long and 1/8 inch in diameter, is applied to the apex of the cone for one minute.

4.4.2.10 Spontaneous heating. The apparatus for the spontaneous heating test consists essentially of a vertical, cylindrical test chamber, 7 inches in height and 4 inches in diameter, surrounded at the side and bottom by a water bath heated by an electric hot plate. The test chamber is provided with a cover having two air vent tubes and an opening for introduction of a thermocouple or a maximum registering thermometer. The sample is contained within the test chamber by a cylinder of wire gauze, 6 inches in height and 1-1/2 inches in diameter. This cylinder is concentric with the axis of the test chamber, there being an annular space through air of 1-1/4 inches between the cylinder and the wall of the chamber.

In conducting the spontaneous heating test, the containing cylinder is filled with the product and placed within the test chamber. The thermocouple or thermometer is inserted in the product so as to measure the temperature at the approximate center of the sample. The cover is then placed on the test chamber, and the water bath maintained at boiling for a period of 72 hours.

A rise in the temperature of the sample above the temperature of the ambient water bath under the conditions of the test is an indication of spontaneous heating of the product.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking. The specified bulk quantity (see 6.2) shall be packed in the applicable type and level (see 6.2) of container specified in PPP-P-45 and marked in accordance with PPP-P-45.

6. NOTES

6.1 Intended use. The sweeping compounds covered by this specification are intended for use in the maintenance of floors.

Type I is suitable for use on wood, steel, and concrete floors, and flooring not affected by sand or mineral oil.

Type II is suitable for use on wood, steel, and concrete floors and flooring not affected by sand.

Type III is suitable for use on wood, ceramic, asphalt and resilient floors.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Selection of the type and level of packing (see 5.1).
(d) Selection of size of packing container (see 5.1).

MILITARY INTEREST:

DOD has waived coordination on revisions and amendments to this Federal Specification until further notice.

CIVIL AGENCY COORDINATING ACTIVITIES:

CPSC

PREPARING ACTIVITY:

GSA-FSS

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